

Tables and PERFORM..VARYING



PROCEDURE DIVISION.

Begin.

OPEN INPUT TaxFile

READ TaxFile

AT END SET EndOfTaxFile TO TRUE

END-READ

PERFORM UNTIL EndOfTaxFile

ADD TaxPaid TO TaxTotal

READ TaxFile

AT END SET EndOfTaxFile TO TRUE

END-READ

END-PERFORM.

DISPLAY "Total taxes are ", TaxTotal

CLOSE TaxFile

STOP RUN.

The program to calculate the total taxes paid for the country is easy to write.

BUT.

What do we do if we want to calculate the taxes paid in each **county**?



PROCEDURE DIVISION.

Begin.

OPEN INPUT TaxFile

READ TaxFile

AT END SET EndOfTaxFile TO TRUE

END-READ

PERFORM SumCountyTaxes UNTIL EndOfTaxFile

DISPLAY "County 1 total is ", County1TaxTotal

: : : **24 Statements** : : :

DISPLAY "County 26 total is ", County26TaxTotal

CLOSE TaxFile

STOP RUN.

SumCountyTaxes.

IF CountyNum = 1 ADD TaxPaid TO County1TaxTotal

END-IF

: : : **24 Statements** : : :

IF CountyNum = 26 ADD TaxPaid TO County26TaxTotal

END-IF

READ TaxFile

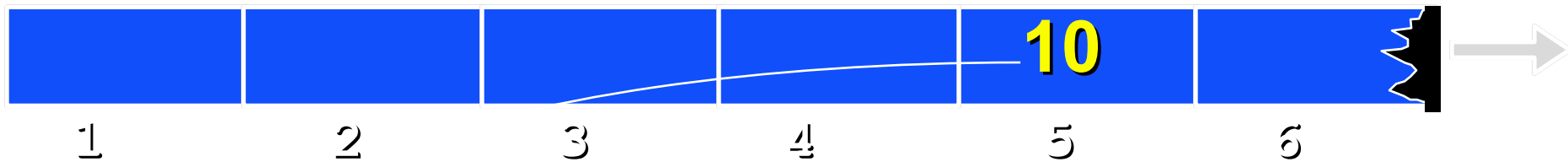
AT END SET EndOfTaxFile TO TRUE

END-READ

58 Statements

Tables/Arrays

A table is a contiguous sequence of memory locations called **elements**, which all have the **same name**, and are uniquely identified by that name and by their **position** in the sequence.



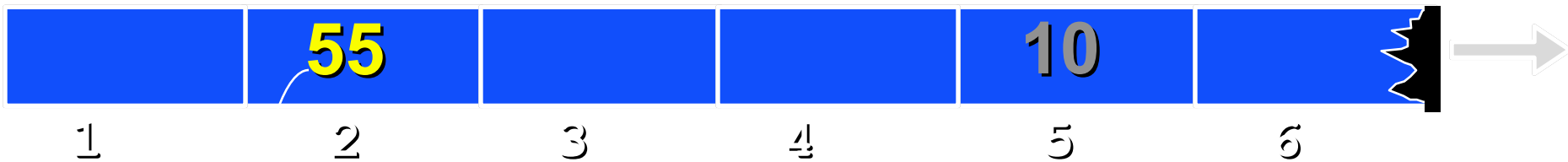
```
MOVE 10 TO CountyTax(5)
```

```
ADD TaxPaid TO CountyTax(CountyNum)
```

```
ADD TaxPaid TO CountyTax(CountyNum + 2)
```

Tables/Arrays

A table is a contiguous sequence of memory locations called **elements**, which all have the **same name**, and are uniquely identified by that name and by their **position** in the sequence.



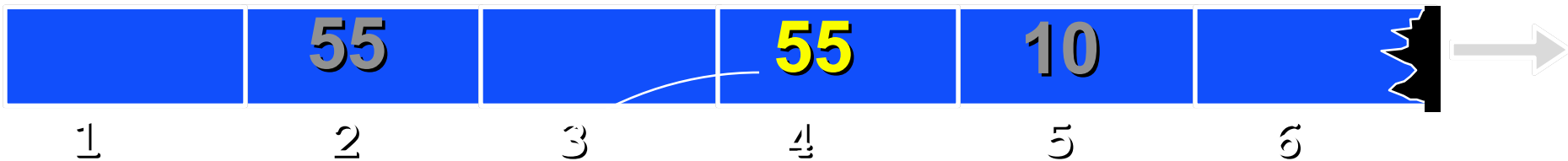
```
MOVE 10 TO CountyTax(5)
```

```
ADD TaxPaid TO CountyTax(CountyNum)
```

```
ADD TaxPaid TO CountyTax(CountyNum + 2)
```

Tables/Arrays

A table is a contiguous sequence of memory locations called **elements**, which all have the **same name**, and are uniquely identified by that name and by their **position** in the sequence.



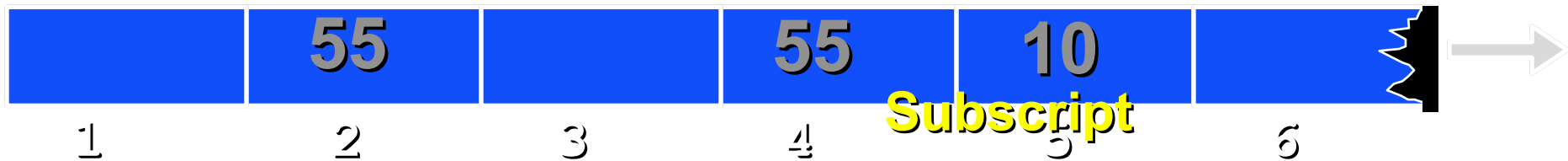
```
MOVE 10 TO CountyTax(5)
```

```
ADD TaxPaid TO CountyTax(CountyNum)
```

```
ADD TaxPaid55 TO CountyTax(CountyNum2 + 2)
```

Tables/Arrays

A table is a contiguous sequence of memory locations called **elements**, which all have the **same name**, and are uniquely identified by that name and by their **position** in the sequence. The position index is called a **subscript**.



MOVE 10 TO CountyTax (**5**)

ADD TaxPaid TO CountyTax (**CountyNum**)

ADD TaxPaid TO CountyTax (**CountyNum + 2**)



1

2

3

4

5

6

PROCEDURE DIVISION.

Begin.

OPEN INPUT TaxFile

READ TaxFile

AT END SET EndOfTaxFile TO TRUE

END-READ

PERFORM UNTIL EndOfTaxFile

ADD TaxPaid TO CountyTax(CountyNum)

READ TaxFile

AT END SET EndOfTaxFile TO TRUE

END-READ

END-PERFORM.

PERFORM VARYING Idx FROM 1 BY 1

UNTIL Idx GREATER THAN 26

DISPLAY "County ", CountyNum

" tax total is " CountyTax(Idx)

END-PERFORM

CLOSE TaxFile

STOP RUN.

Subscript

9 Statements

A-89432	CLARE	7894.55
---------	-------	---------



```
IF CountyName = "CARLOW"  
    ADD TaxPaid TO CountyTax(1)  
END-IF
```

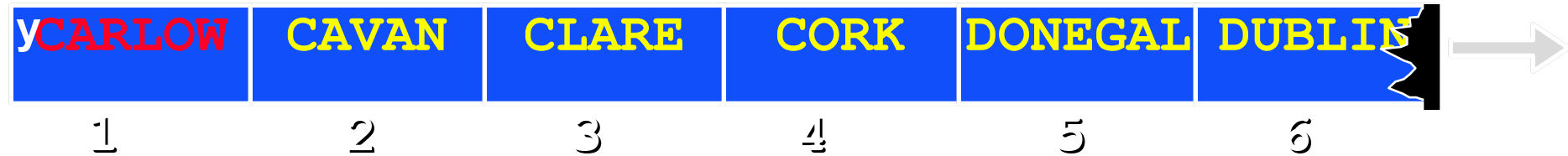
```
IF CountyName = "CAVAN"  
    ADD TaxPaid TO CountyTax(2)  
END-IF
```

```
:      :      :      :      :  
:      :      :      :      :
```

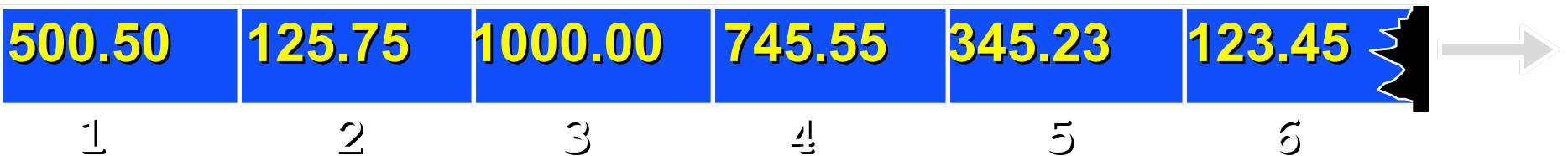
24 TIMES

A-89432	CLARE	7894.55	1
---------	-------	---------	---

1	2	3	4	5	6
YCARLOW	CAVAN	CLARE	CORK	DONEGAL	DUBLIN



1	2	3	4	5	6
500.50	125.75	1000.00	745.55	345.23	123.45



```
PERFORM VARYING Idx FROM 1 BY 1
      UNTIL County(Idx) = CountyName
END-PERFORM
ADD TaxPaid TO CountyTax(Idx)
```

A-89432	CLARE	7894.55	2
---------	-------	---------	---

1	2	3	4	5	6
YCARLOW	CAVAN	CLARE	CORK	DONEGAL	DUBLIN



1	2	3	4	5	6
500.50	125.75	1000.00	745.55	345.23	123.45



```
PERFORM VARYING Idx FROM 1 BY 1
      UNTIL County(Idx) = CountyName
END-PERFORM
ADD TaxPaid TO CountyTax(Idx)
```

A-89432	CLARE	7894.55	3
---------	-------	---------	---

1	2	3	4	5	6
YCARLOW	CAVAN	CLARE	CORK	DONEGAL	DUBLIN



1	2	3	4	5	6
500.50	125.75	1000.00	745.55	345.23	123.45



```
PERFORM VARYING Idx FROM 1 BY 1
      UNTIL County(Idx) = CountyName
END-PERFORM
ADD TaxPaid TO CountyTax(Idx)
```

A-89432	CLARE	7894.55	3
---------	-------	---------	---

YCARLOW	CAVAN	CLARE	CORK	DONEGAL	DUBLIN
1	2	3	4	5	6

500.50	125.75	8894.55	745.55	345.23	123.45
1	2	3	4	5	6

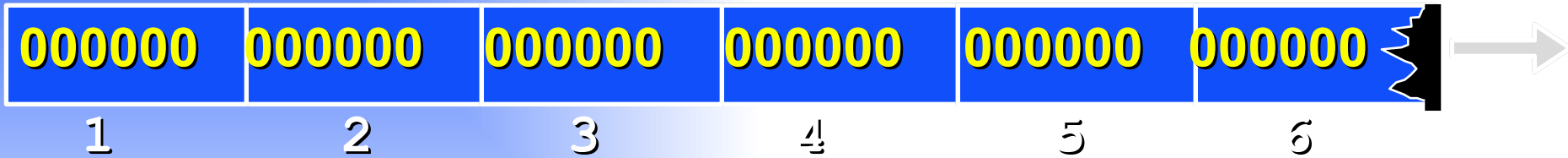
```

PERFORM VARYING Idx FROM 1 BY 1
    UNTIL County(Idx) = CountyName
END-PERFORM
ADD TaxPaid TO CountyTax(Idx)

```

Declaring Tables.

TaxTotals
CountyTax



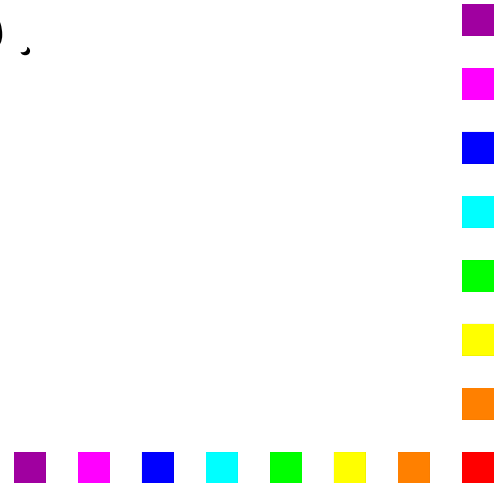
```
01  TaxTotals.  
    02  CountyTax  PIC 9(10)V99  
                  OCCURS 26 TIMES.
```

or

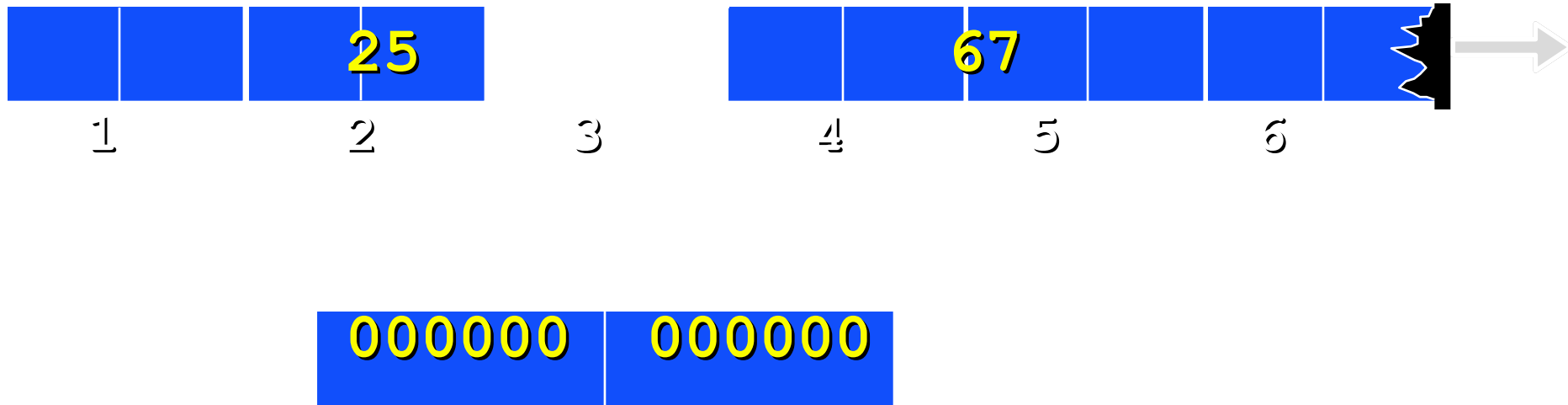
```
02  CountyTax  OCCURS 26 TIMES  
              PIC 9(10)V99.
```

e.g.

```
MOVE ZEROS TO TaxTotals.  
MOVE 20 TO CountyTax(5).
```



Group Items as Elements.



```
01  TaxTotals.  
    02  CountyTaxDetails OCCURS 26 TIMES.  
        03  CountyTax      PIC 9(10)V99.  
        03  PayerCount     PIC 9(7).
```

```
e.g.  MOVE 25 TO PayerCount(2).  
      MOVE 67 TO CountyTax(5).  
      MOVE ZEROS TO CountyTaxDetails(3).
```

PERFORM..VARYING Syntax

PERFORM $\left[\text{1stProc} \left[\left\{ \begin{array}{c} \underline{\text{THRU}} \\ \underline{\text{THROUGH}} \end{array} \right\} \text{EndProc} \right] \right] \left[\text{WITH } \underline{\text{TEST}} \left\{ \begin{array}{c} \underline{\text{BEFORE}} \\ \underline{\text{AFTER}} \end{array} \right\} \right]$

VARYING $\left\{ \begin{array}{c} \text{Identifier1} \\ \text{IndexName1} \end{array} \right\} \underline{\text{FROM}} \left\{ \begin{array}{c} \text{Identifier2} \\ \text{IndexName2} \\ \text{Literal} \end{array} \right\}$

BY $\left\{ \begin{array}{c} \text{Identifier3} \\ \text{Literal} \end{array} \right\} \underline{\text{UNTIL}} \text{Condition1}$

$\left[\begin{array}{c} \underline{\text{AFTER}} \left\{ \begin{array}{c} \text{Identifier4} \\ \text{IndexName3} \end{array} \right\} \underline{\text{FROM}} \left\{ \begin{array}{c} \text{Identifier5} \\ \text{IndexName4} \\ \text{Literal} \end{array} \right\} \\ \underline{\text{BY}} \left\{ \begin{array}{c} \text{Identifier6} \\ \text{Literal} \end{array} \right\} \underline{\text{UNTIL}} \text{Condition2} \end{array} \right] \dots$

$\left[\text{StatementBlock } \underline{\text{END - PERFORM}} \right]$




```
PERFORM VARYING Idx1 FROM 1 BY 1 UNTIL  
    Idx1 EQUAL TO 3  
    DISPLAY Idx1  
END-PERFORM.
```

Move 1 to Idx1

1

Idx1 = 3

Next Statement

Loop Body

Inc Idx1

```
PERFORM VARYING Idx1 FROM 1 BY 1 UNTIL  
    Idx1 EQUAL TO 3  
    DISPLAY Idx1  
END-PERFORM.
```

Move 1 to Idx1

1

Idx1 = 3

Next Statement

Loop Body

Inc Idx1

```
PERFORM VARYING Idx1 FROM 1 BY 1 UNTIL  
    Idx1 EQUAL TO 3  
    DISPLAY Idx1  
END-PERFORM.
```

Move 1 to Idx1

1

Idx1 = 3

Next Statement

Loop Body

Inc Idx1

1

```
PERFORM VARYING Idx1 FROM 1 BY 1 UNTIL  
    Idx1 EQUAL TO 3  
    DISPLAY Idx1  
END-PERFORM.
```

Move 1 to Idx1

2

Idx1 = 3

Next Statement

Loop Body

Inc Idx1

1

```
PERFORM VARYING Idx1 FROM 1 BY 1 UNTIL  
    Idx1 EQUAL TO 3  
    DISPLAY Idx1  
END-PERFORM.
```

Move 1 to Idx1

2

Idx1 = 3

Next Statement

Loop Body

Inc Idx1

1

```
PERFORM VARYING Idx1 FROM 1 BY 1 UNTIL  
    Idx1 EQUAL TO 3  
    DISPLAY Idx1  
END-PERFORM.
```

Move 1 to Idx1

2

Idx1 = 3

Next Statement

Loop Body

Inc Idx1

1
2

```
PERFORM VARYING Idx1 FROM 1 BY 1 UNTIL  
    Idx1 EQUAL TO 3  
    DISPLAY Idx1  
END-PERFORM.
```

Move 1 to Idx1

3

Idx1 = 3

Next Statement

Loop Body

Inc Idx1

1
2

```
PERFORM VARYING Idx1 FROM 1 BY 1 UNTIL  
    Idx1 EQUAL TO 3  
    DISPLAY Idx1  
END-PERFORM.
```

Move 1 to Idx1

3

Idx1 = 3

Next Statement

Loop Body

Inc Idx1

1
2
Exit value = 3


```
PERFORM IterationCount VARYING Idx1 FROM 1 BY 2
                        UNTIL   Idx1 EQUAL TO 5
                        AFTER    Idx2 FROM 6 BY -1
                        UNTIL    Idx2 LESS THAN 4
```

Move 1 to Idx1
Move 6 to Idx2

Idx1 = 5

Next Statement

Idx2 < 4

IterationCount

Move 6 to Idx2
Inc Idx1

Dec Idx2

T	Idx1	Idx2
1	1	6
2	1	5
3	1	4
4	3	6
5	3	5
6	3	4
X	= 5	= 6